JEFF MUTUKU
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Human Performance Researcher |Biomechanics| R&D Analytics

PROFESSIONAL SUMMARY

Human performance researcher specializing in standardized human-subject testing, protocol execution, and multimodal data synthesis. Combines extensive laboratory and athletic testing experience with AI-assisted analytical workflows to translate complex participant feedback, biomechanical streams, and performance data into clear, actionable product insights.

EDUCATION

California Baptist University
Master of Science in Kinesiology (Concentration: Exercise Science)
Academic Record: Cumulative GPA: 3.89 / 4.00

Notre Dame College
Bachelor of Arts in Sports Management
Academic Record: Cumulative GPA: 3.70 / 4.00 **|** **Honors:** Cum Laude, Dean's List
Athletic Distinction: NCAA Division I Athletic Scholarship Recipient

PROFESSIONAL EXPERIENCE

Performance Scientist **May 2026 - Current**
Wellth Performance

- Conduct 2+ metabolic assessments per week, including resting metabolic rate (RMR) and maximal exercise testing, evaluating oxygen consumption, ventilatory responses, and aerobic capacity.
- Perform DEXA and 3D body-composition assessments for 35+ clients, evaluating fat mass, lean mass, and regional composition to support individualized recommendations and progress monitoring.
- Develop and apply a reusable AI-assisted metabolic analysis and reporting workflow that processes VO_2max testing data, applies predefined physiological criteria to identify candidate ventilatory thresholds and estimate training zones, and generates standardized reports for performance staff.
- Initiated a DEXA research and analytics project to organize longitudinal body-composition data and investigate associations between scan frequency, client engagement, and changes in body composition.

Intern, Performance Coach **Jul 2025 - May 2026**
Athletic Performance Center, California Baptist University

- Supported performance operations across 12 NCAA Division I athletic programs, managing athlete diagnostics and load monitoring pipelines.
- Executed force-plate countermovement jump (CMJ) assessments to analyze ground reaction forces, rate of force development (RFD), impulse, and fatigue status.
- Tracked external workload kinetics using STATSports GPS wearable sensors and accelerometers to inform training volume adjustments during conference championship runs.
- Worked alongside strength and conditioning staff and sport-specific coaches within a multidisciplinary performance environment.

RESEARCH EXPERIENCE

Performance Researcher – Applied Biomechanics & Human Performance
Post-Activation Potentiation & Sprint Swimming Performance in NCAA Division I Athletes

- Executed an IRB-approved study involving 12 NCAA Division I swimmers and 72 experimental trials to examine acute performance responses.
- Quantified takeoff kinetics (Takeoff Power in Watts) and sprint kinematics (25-yard time in seconds) using ankle-mounted accelerometers and digital timing systems.
- Programmed an interactive, client-facing web dashboard using HTML5, CSS3, JavaScript, and Plotly.js, deployed via Vercel, enabling real-time toggling between group-level KPIs and individual subject response profiles.

Co-Principal Investigator – Human Subjects Research
Seasonal Variations of Heart Rate Recovery in NCAA Division I Swimmers

- Co-led longitudinal research investigating seasonal variations in Heart Rate Recovery (HRR) in 11 NCAA Division I swimmers during in-season vs. off-season conditions.
- Administered Bruce Protocol treadmill tests to volitional fatigue and logged continuous 5-minute post-exercise recovery heart rates at 5-second intervals.
- Modeled autonomic recovery kinetics in Python using single-exponential decay equations [$Y = Y0 + A \cdot e^{-t/\tau}$] to quantify parasympathetic reactivation via the time constant Tau (τ).
- Identified a statistically significant increase in off-season Tau (90.8 ± 8.2s) vs. in-season (85.6 ± 6.9s), quantifying off-season aerobic decay and guiding intervention design.

TECHNICAL EXPERTISE

- **Human-Subject Research Protocols:** IRB Protocol Execution, Participant Screening, Informed Consent, Testing Standardization, Quality Control & Data Integrity.
- **Laboratory & Field Instrumentation:** DEXA Body Composition, 3D Body Scanning, Indirect Calorimetry (VO_2max, RMR), Force Plates (CMJ Analytics), STATSports GPS, Accelerometry.
- **Data Science & Quantitative Modeling:** Python (Pandas, NumPy, SciPy), Time-Series Analysis, Exponential Decay Modeling, Non-Linear Curve Fitting, Exploratory Data Analysis.
- **Software & Analytical Infrastructure:** Plotly, Streamlit, Power BI, Advanced Excel, JavaScript, HTML5/CSS3, Git/GitHub, Vercel Web Deployment.

CERTIFICATIONS & AWARDS

- **Licensure & Certifications:** California Department of Public Health X-Ray Technician Permit, STATSports Certified GPS User, FIFA Diploma in Football Medicine, NASM-CPT.
- **Professional Affiliations:** NSCA Member, Certified Performance and Sport Scientist (CPSS) Exam Candidate (2026).
- **Distinctions:** Former Division I Athlete, Scholastic All-American, Junior Black Belt in Taekwondo.